

Combined Effects of Sex & Exercise Habits on Body Weight

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1. CORE RESEARCH OBJECTIVE

This academic project investigates the predictive powers and interconnected relationship between a physiological demographic variable (Biological Sex) and a lifestyle behavioral habit (Exercise Frequency) on overall adult body mass.

- Phase I (Baseline Inherent Gap):** Does biological sex define a massive, highly significant baseline difference in overall standard body weight parameters?
- Phase II (Interaction Matrix):** Does structural physical activity influence weight reduction uniformly across genders, or do the two variables create an interdependent interference pattern (Interaction Effect)?

2. VARIABLE OPERATIONAL FRAMEWORK

To avoid structural biases and satisfy rigorous statistical rules, data cells are perfectly balanced (n = 250 per matrix group).

Variable Type	Variable Name	Scale Type
Independent 1	Sex_Gender	Categorical [Male / Female]
Independent 2	Exercise_Habit	Categorical [High / Low]
Dependent Metric	Body_Weight_KG	Continuous Numerical (KG)

3. PHASE I: MACRO BASELINE DIVERGENCE

Omitted from lifestyle choices, biological sex exerts an overriding influence on body weight. Mean averages showcase clear macro-level tracking:

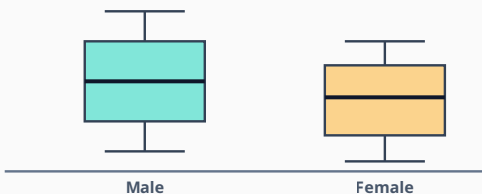
74.38 kg

Male General Mean

62.60 kg

Female General Mean

FIGURE 1: BASELINE DISTRIBUTION GAP BLOCK



4. PHASE II: STRATIFIED SUBGROUPS

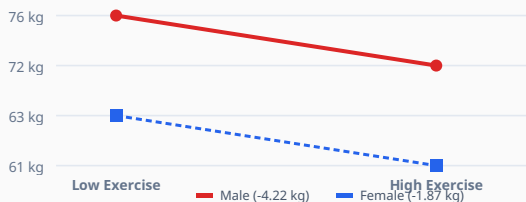
Analyzing cross-sections shows profound variance in weight behavior when physical habits interact with biological realities.

Cohort	Exercise	Mean Weight	SD	Net Variance Outcome
Female	Low Habit	63.53 kg	3.96	Active females are 1.87 kg lighter on average.
	High Habit	61.66 kg	3.35	
Male	Low Habit	76.49 kg	4.83	Active males are 4.22 kg lighter on average.
	High Habit	72.27 kg	4.49	

5. INFERENTIAL EVIDENCE & SLOPES

A Simple Main Effects evaluation confirms that exercise impacts weight dynamics between groups with vastly asymmetrical strengths.

FIGURE 2: NON-PARALLEL INTERACTION SLOPES



- Within-Male Impact:** Powerful statistical impact registered ($F = 102.12$, $p < 0.001$). Exercise triggers massive physiological downshifts.
- Within-Female Impact:** Highly statistically valid ($F = 32.56$, $p < 0.001$), yet records a noticeably gentler reduction coefficient compared to men.

6. FINAL CORE CONCLUSIONS

- Universal Success Metric:** Regular structured exercise routine significantly lowers overall body weight indicators across all demographic splits ($p < 0.001$).
- Disproportionate Return:** Exercise impact vectors act more aggressively within males, resulting in double the structural loss rate compared to females (4.22 kg vs 1.87 kg).
- True Interaction Confirmed:** Non-parallel lines in trend mapping mathematically establish an explicit interaction effect ($p = 1.12e-05$). Sex and lifestyle do not work independently; they cross-interfere to scale final body mass outcomes.